

Smart Hospital Patient Management System (Java OOP Project)

Course: Object Oriented Programming (CSE 282)
Submission Ready Report Template

Abstract

This project develops a Smart Hospital Patient Management System using Java. The system demonstrates Encapsulation, Inheritance, Polymorphism and Abstraction.

Introduction

Hospitals require efficient management of patients, doctors, nurses and pharmacists. This project models these entities using OOP concepts.

Problem Analysis

Manual record keeping causes inefficiency, duplication and data loss. A computerized solution improves reliability.

Stakeholder Analysis

Doctors need diagnosis management. Nurses need patient care logs. Pharmacists need medication records. Patients need accurate information management.

Functional Requirements

Add patient, update patient, assign patient, generate reports, maintain medication history, manage staff.

Non Functional Requirements

Security, maintainability, usability, reliability, scalability and performance.

OOP Mapping

Encapsulation through Patient class. Inheritance through HospitalStaff hierarchy. Polymorphism through generateReport(). Abstraction through MedicalService interface.

Class Design

Patient, MedicalService, HospitalStaff, Doctor, Nurse, Pharmacist and HospitalManagement classes.

Encapsulation

Patient fields are private and accessed through getters and setters.

Inheritance

Doctor, Nurse and Pharmacist inherit from HospitalStaff.

Polymorphism

Each subclass overrides generateReport() with different outputs.

Abstraction

MedicalService declares assignPatient(), performDuty() and generateReport().

IDE Selection

IntelliJ IDEA is recommended because of debugging support, code completion and project management features.

UML Diagram

MedicalService -> HospitalStaff -> Doctor/Nurse/Pharmacist. Patient is associated with staff members.

Testing

Create sample patients and assign them to staff. Verify role-specific report generation.

Sample Test Cases

Add Patient, Assign Patient, Generate Doctor Report, Generate Nurse Report, Generate Pharmacist Report.

Results

The system successfully demonstrates all four pillars of OOP.

Challenges

Designing class hierarchy, ensuring proper abstraction and implementing polymorphism.

Future Scope

GUI version, database integration, appointment scheduling and billing system.

Conclusion

The project successfully applies Java OOP principles to a real-world hospital management scenario.

Submission Checklist

1. Add your name, ID, section and semester.
2. Paste screenshots of IntelliJ/VS Code project structure.
3. Paste screenshots of code execution/output.
4. Include UML diagram image.
5. Attach Java source files in a ZIP folder during submission.